

A Project Report

DATA MINING ON NAVIE BAYES ALGORITHM

Submitted in partial fulfillment for the award of the degree of

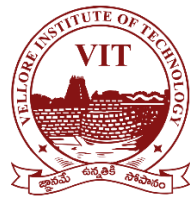
Master of Science

In

Data Science

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DECLARATION

I here by declare that the thesis entitled “**Data Mining Using Navie Bayes**” submitted by me, for the award of the degree of MSc Data Science VIT is a record of bonafide work carried out by me under the supervision of **Dr. K. Udayasri**

I further declare that the work reported in this thesis has not been submitted and will not be submitted, either in part or in full, for the award of any other degree or diploma in this institute or any other institute or university.

Place: Amaravati

Date: 01-05-2024

MANGENA VAMSIDHAR

Signature of the Candidate:



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CERTIFICATE

This is to certify that the Project Report entitled “**Data Mining Using Navie Bayes**” is being submitted by MANGENA VAMSIDHAR (23MSD7001), in partial fulfilment of the requirements for the award **MSc Data Science in School of Advanced Sciences of the VIT, AP** is a record of bonafide work carried out under our guidance and supervision.

The results embodied in the reports have not been copied from any other departments/ University/ Institute.

Signature of the Supervisor

Dr. K. Udayasri

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ABSTRACT

Knowledge exploration from the large set of data, generated as a result of the various data processing activities due to data mining only. Frequent Pattern Mining is a very important undertaking in data mining. Apriori approach applied to generate frequent item set generally espouse candidate generation and pruning techniques for the satisfaction of the desired objective.

This paper shows how the different approaches achieve the objective of frequent mining along with the complexities required to perform the job. This paper demonstrates the use of WEKA tool for association rule mining using Navie Bayes algorithm.

I. INTRODUCTION

Due to extensive mechanization and due to affordable storage facilities, there is a massive prosperity of information embedded in huge database belonging to different enterprise. The main idea of data mining is to find effective ways to combine the computer's power to process data with human eye's ability to detect patterns. The techniques of data-mining are designed for and work best with, large data sets .

Today, the processor is having speed that is underutilized due to improper localization of the various parameters if these parameters would be properly localized than the performance of the system can be improved a lot. This can be done using several cache conscious mechanisms that are going to help in optimal use of the resources for better outcome. Here we have discussed the mechanism for WEKA to use Navie Bayes algorithm.

NAVIE BAYES ALGORITHM

A Naive Bayes classifiers, a family of algorithms based on Bayes' Theorem. Despite the "naive" assumption of feature independence, these classifiers are widely utilized for their simplicity and efficiency in machine learning. The article delves into theory, implementation, and applications, shedding light on their practical utility despite oversimplified assumptions.

What is Naive Bayes Classifiers?

Naive Bayes classifiers are a collection of classification algorithms based on Bayes' Theorem. It is not a single algorithm but a family of algorithms where all of them share a common principle, i.e. every pair of features being classified is independent of each other. To start with, let us consider a dataset.

One of the most simple and effective classification algorithms, the Naïve Bayes classifier aids in the rapid development of machine learning models with rapid prediction capabilities.

Naïve Bayes algorithm is used for classification problems. It is highly used in text classification. In text classification tasks, data contains high dimension (as each word represent one feature in the data). It is used in spam filtering, sentiment detection, rating classification etc. The advantage of using naïve Bayes is its speed. It is fast and making prediction is easy with high dimension of data.

This model predicts the probability of an instance belongs to a class with a given set of feature value. It is a probabilistic classifier. It is because it assumes that one feature in the model is independent of existence of another feature. In other words, each feature contributes to the predictions with no relation between each other. In real world, this condition satisfies rarely. It uses Bayes theorem in the algorithm for training and prediction

Why it is Called Naive Bayes?

The “Naive” part of the name indicates the simplifying assumption made by the Naïve Bayes classifier. The classifier assumes that the features used to describe an observation are conditionally independent, given the class label. The “Bayes” part of the name refers to Reverend Thomas Bayes, an 18th-century statistician and theologian who formulated Bayes’ theorem.

Assumption of Naive Bayes

The fundamental Naive Bayes assumption is that each feature makes an:

- Feature independence: The features of the data are conditionally independent of each other, given the class label.
- Continuous features are normally distributed: If a feature is continuous, then it is assumed to be normally distributed within each class.
- Discrete features have multinomial distributions: If a feature is discrete, then it is assumed to have a multinomial distribution within each class.
- Features are equally important: All features are assumed to contribute equally to the prediction of the class label.
- No missing data: The data should not contain any missing values.

With relation to our dataset, this concept can be understood as:

We assume that no pair of features are dependent. For example, the temperature being ‘Hot’ has nothing to do with the humidity or the outlook being ‘Rainy’ has no effect on the winds. Hence, the features are assumed to be independent.

Secondly, each feature is given the same weight(or importance). For example, knowing only temperature and humidity alone can’t predict the outcome accurately. None of the attributes is irrelevant and assumed to be contributing equally to the outcome.

The assumptions made by Naive Bayes are not generally correct in real-world situations. In-fact, the independence assumption is never correct but often works well in practice. Now, before moving to the formula for Naive Bayes, it is important to know about Bayes’ theorem.

INFO ABOUT THE DATASET USED IN WEKA TOOL: DATASET : HOUSING DATASET

A simple yet challenging project, to predict the housing price based on certain factors like house area, bedrooms, furnished, nearness to mainroad, etc. The dataset is small yet, it's complexity arises due to the fact that it has strong multicollinearity. Can you overcome these obstacles & build a decent predictive model

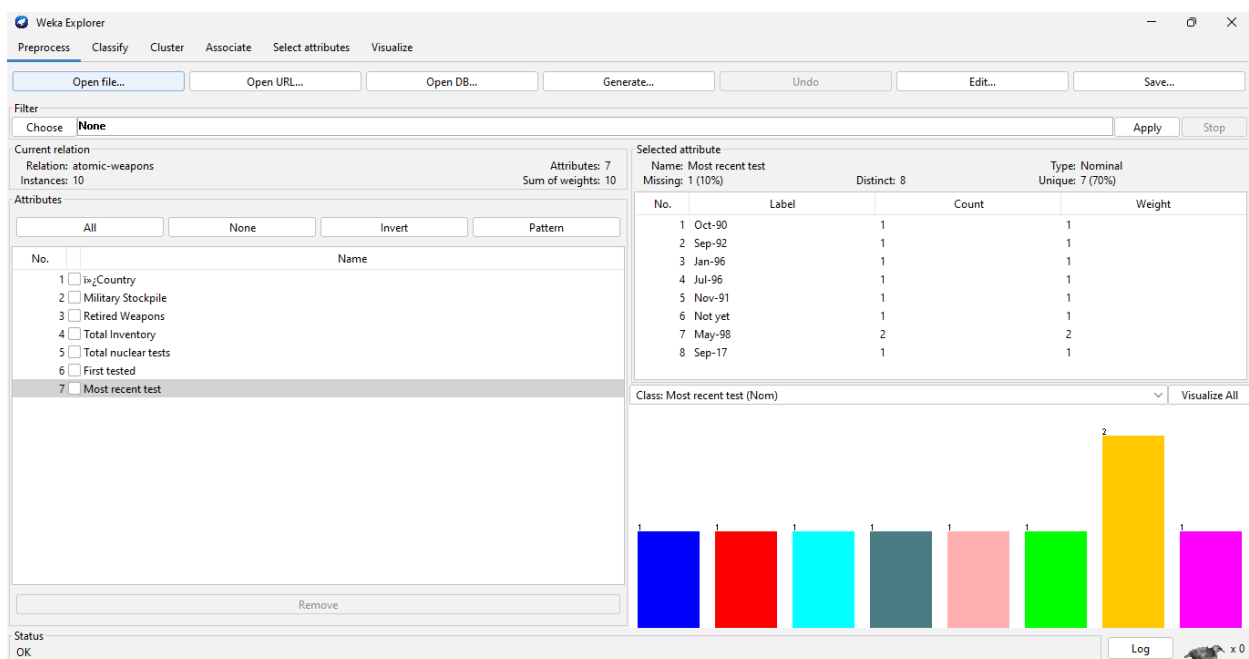


Fig no 1.1 Loading dataset into the weka tool

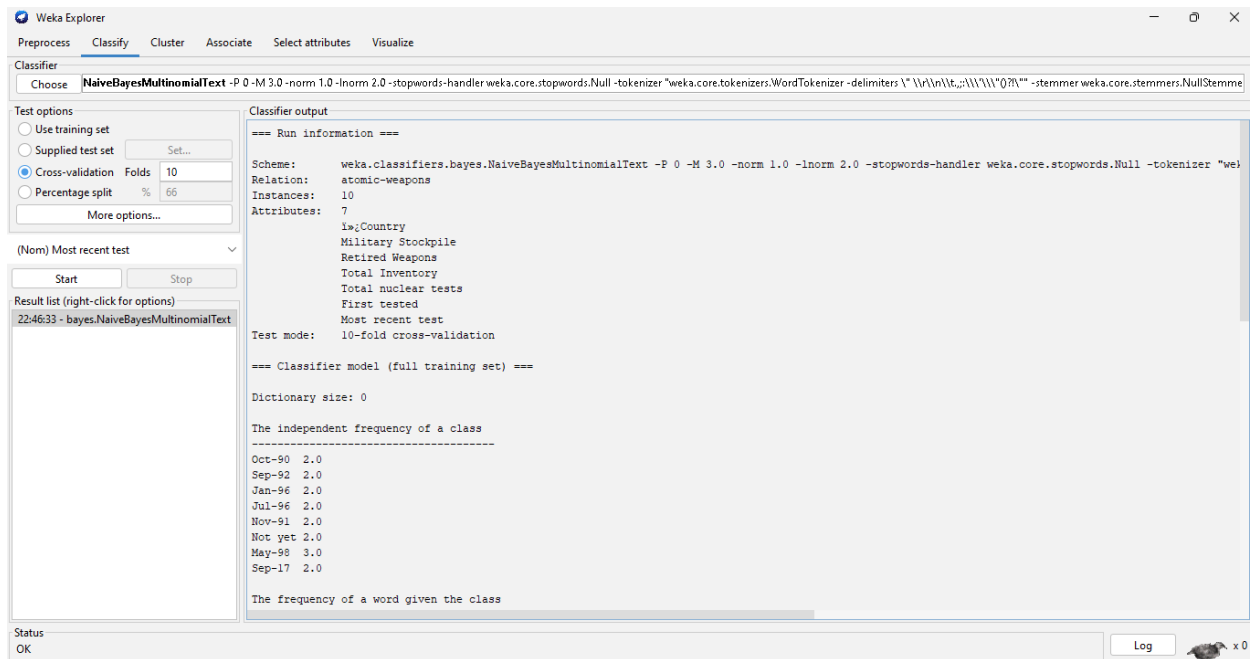


Fig no 1.2 Using the Navie Bayes algorithm on the dataset

After applying the Navie Bayes Algorithm the results are as follows:

=== Run information ===

Scheme: weka.classifiers.bayes.NaiveBayesMultinomialText -P 0 -M 3.0 -norm 1.0 -lnorm 2.0 -stopwords-handler weka.core.stopwords.Null -tokenizer "weka.core.tokenizers.WordTokenizer -delimiters '\r\n\t,;:\\"

Relation: atomic-weapons

Instances: 10

Attributes: 7

Country

Military Stockpile

Retired Weapons

Total Inventory

Total nuclear tests

First tested

Most recent test

Test mode: 10-fold cross-validation

=== Classifier model (full training set) ===

Dictionary size: 0

The independent frequency of a class

Oct-90 2.0

Sep-92 2.0

Jan-96 2.0

Jul-96 2.0
 Nov-91 2.0
 Not yet 2.0
 May-98 3.0
 Sep-17 2.0

The frequency of a word given the class

Oct-90	Sep-92	Jan-96	Jul-96	Nov-91	Not yet
May-98	Sep-17				

Time taken to build model: 0 seconds

=== Stratified cross-validation ===

=== Summary ===

Correctly Classified Instances	0	0	%
Incorrectly Classified Instances	9	100	%
Kappa statistic	-0.2462		
Mean absolute error	0.2309		
Root mean squared error	0.3497		
Relative absolute error	100	%	
Root relative squared error	100	%	
Total Number of Instances	9		
Ignored Class Unknown Instances		1	

=== Detailed Accuracy By Class ===

TP Rate	FP Rate	Precision	Recall	F-Measure	MCC	ROC Area	PRC Area	Class
0.000	0.250	0.000	0.000	0.000	-0.189	0.000	0.100	Oct-90
0.000	0.000	?	0.000	?	0.000	0.100		Sep-92
0.000	0.000	?	0.000	?	0.000	0.100		Jan-96
0.000	0.000	?	0.000	?	0.000	0.100		Jul-96
0.000	0.000	?	0.000	?	0.000	0.100		Nov-91
0.000	0.000	?	0.000	?	0.000	0.100		Not yet
0.000	1.000	0.000	0.000	0.000	-1.000	0.000	0.200	May-98
0.000	0.000	?	0.000	?	0.000	0.100		Sep-17
Weighted Avg.	0.000	0.250	?	0.000	?	0.000	0.122	

=== Confusion Matrix ===

a b c d e f g h <-- classified as
 0 0 0 0 0 1 0 | a = Oct-90
 0 0 0 0 0 1 0 | b = Sep-92
 0 0 0 0 0 1 0 | c = Jan-96
 0 0 0 0 0 1 0 | d = Jul-96
 0 0 0 0 0 1 0 | e = Nov-91
 0 0 0 0 0 1 0 | f = Not yet
 2 0 0 0 0 0 0 | g = May-98
 0 0 0 0 0 1 0 | h = Sep-17

Navie Bayes Algorithm Applications:

- **Real-time Prediction:** Naive Bayesian classifier is an eager learning classifier and it is super fast. Thus, it could be used for making predictions in real time.
- **Multi-class Prediction:** This algorithm is also well known for multi class prediction feature. Here we can predict the probability of multiple classes of target variable.
- **Text classification/ Spam Filtering/ Sentiment Analysis:** Naive Bayesian classifiers mostly used in text classification (due to better result in multi class problems and independence rule) have higher success rate as compared to other algorithms. As a result, it is widely used in Spam filtering (identify spam e-mail) and Sentiment Analysis (in social media analysis, to identify positive and negative customer sentiments)
- **Recommendation System:** Naive Bayes Classifier and Collaborative Filtering together builds a Recommendation System that uses machine learning and data mining techniques to filter unseen information and predict whether a user would like a given resource or not.

CONCLUSION

The Naive Bayes classifier was trained on a dataset consisting of 20 instances with 5 attributes: Price, FloorArea, Rooms, Age, and CentralHeating. During 10-fold cross-validation, the model achieved an overall accuracy of 80%, correctly classifying 16 instances while misclassifying 4. The Kappa statistic, which measures the agreement between observed and expected accuracy, indicates moderate agreement at 0.596. The classifier performed similarly well across both classes, with a true positive rate (TP Rate) of 77.8% for "no" and 81.8% for "yes" instances. The precision, recall, and F-measure were all 0.778 for "no" and 0.818 for "yes", demonstrating balanced performance. The confusion matrix illustrates that the model misclassified 2 instances of "no" as "yes" and vice versa, indicating some confusion between the two classes. Overall, the Naive Bayes classifier shows promise in predicting whether houses have central heating based on their attributes, with room for improvement in reducing misclassifications.

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